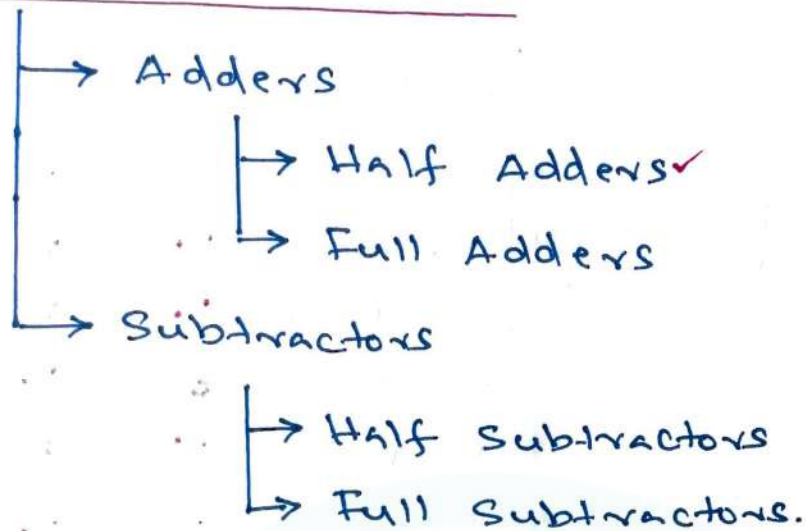
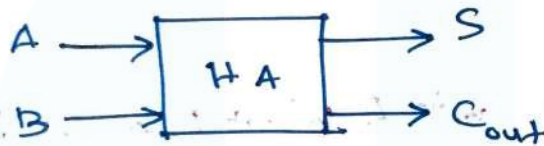


Module 3

Arithmetic Circuits.



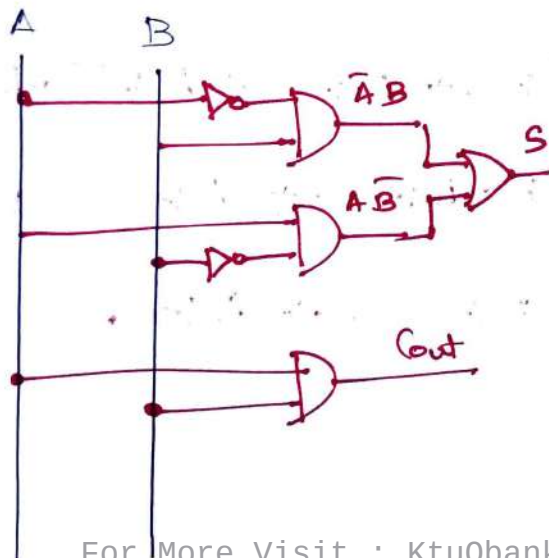
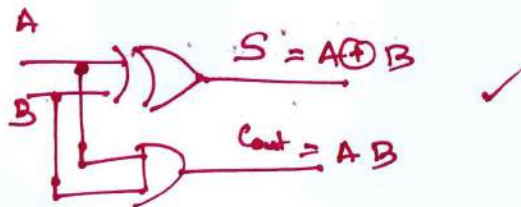
Half Adders.



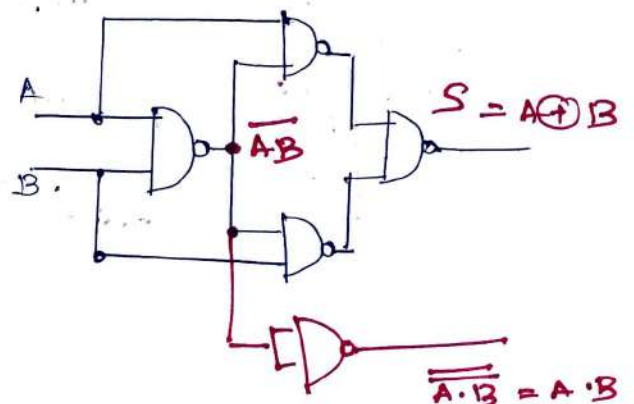
A	B	S	Cout
0	0	0	0
0	1	1✓	0
1	0	1✓	0
1	1	0	1✓

$$S = \bar{A}B + A\bar{B} = A \oplus B$$

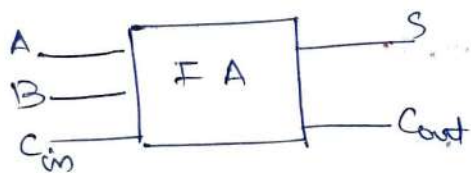
$$C_{out} = \underline{A \cdot B}$$



Using NAND



Full Adder



	A	B	C_{in}	S	C_{out}
0	0	0	0	0	0
1	0	0	1	1 ✓	0
2	0	1	0	1 ✓	0
3	0	1	1	0	1 ✓
4	1	0	0	1 ✓	0
5	1	0	1	0	1 ✓
6	1	1	0	0	1 ✓
7	1	1	1	1 ✓	1 ✓

S

AB \ C	$\bar{C}$	C
$\bar{A}\bar{B}$		①
$\bar{A}B$	①	
$A\bar{B}$		①
AB	①	

$$\begin{aligned}
 S &= \bar{A}\bar{B}C_{in} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC \\
 &= \bar{A}(\bar{B}C_{in} + B\bar{C}) + A(\bar{B}\bar{C} + BC) \\
 &= \bar{A}(B \oplus C) + A(\overline{B \oplus C})
 \end{aligned}$$

$$S = \underline{\underline{A \oplus B \oplus C}}$$

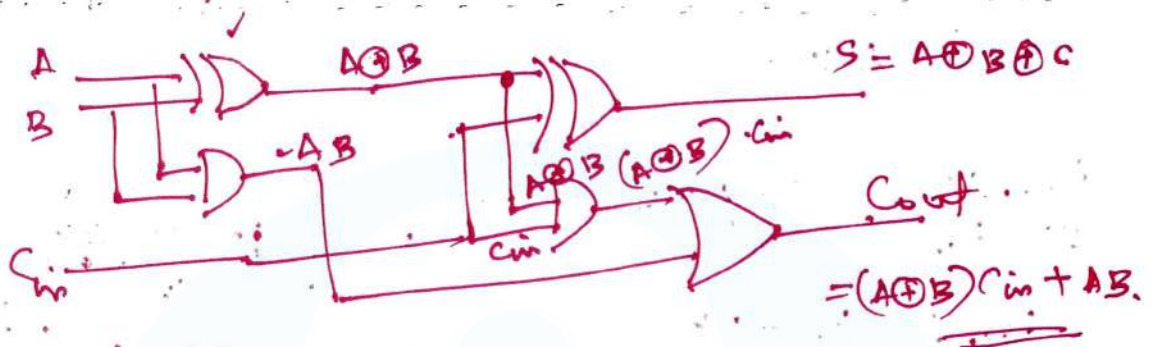
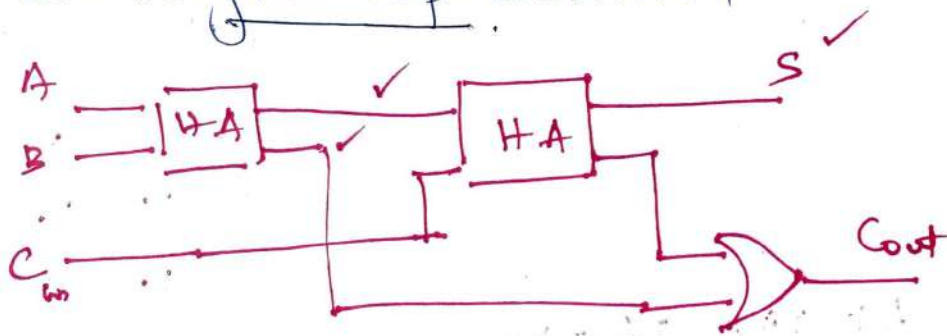
AB \ C	$\bar{C}$	C
$\bar{A}\bar{B}$		
$\bar{A}B$		①
$A\bar{B}$	①	①
AB		①

$$C_{out} = \underline{\underline{AB + BC + AC}} \quad \checkmark$$

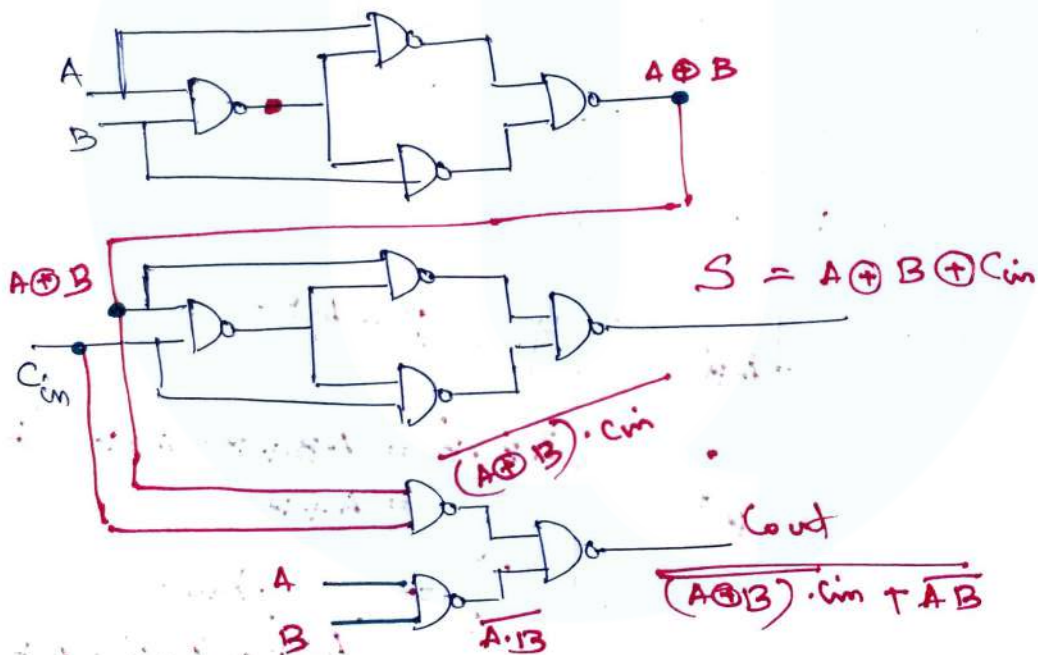
$$\begin{aligned}
 C_{out} &= \bar{A}BC + A\bar{B}C + A\bar{B}\bar{C} + ABC \\
 &= (\bar{A}B + A\bar{B})C + AB(\bar{C} + C)
 \end{aligned}$$

$$C_{out} = \underline{\underline{(A \oplus B)C + AB}} \quad \checkmark$$

Block Diagram Representation



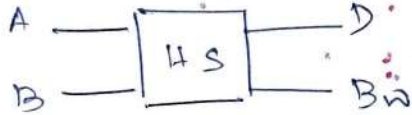
Using NAND gate only.



$$C_{out} = (A \oplus B) C_{in} + A \cdot B$$

$$= ((A \oplus B) \cdot C_{in}) + (A \cdot B)$$

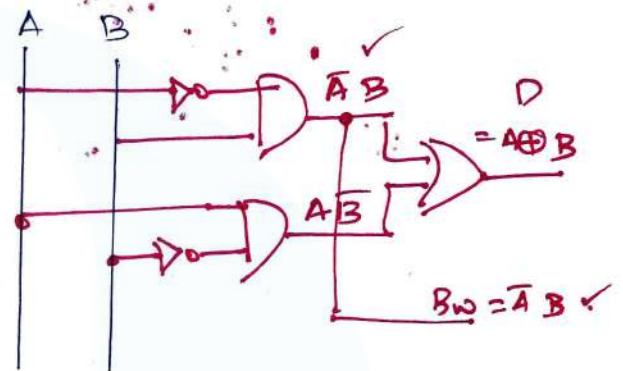
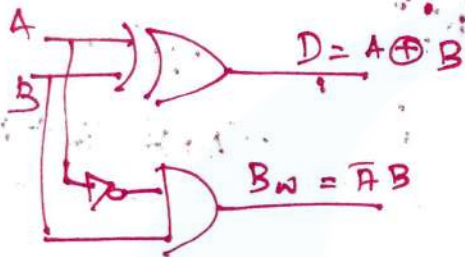
Half Subtractor



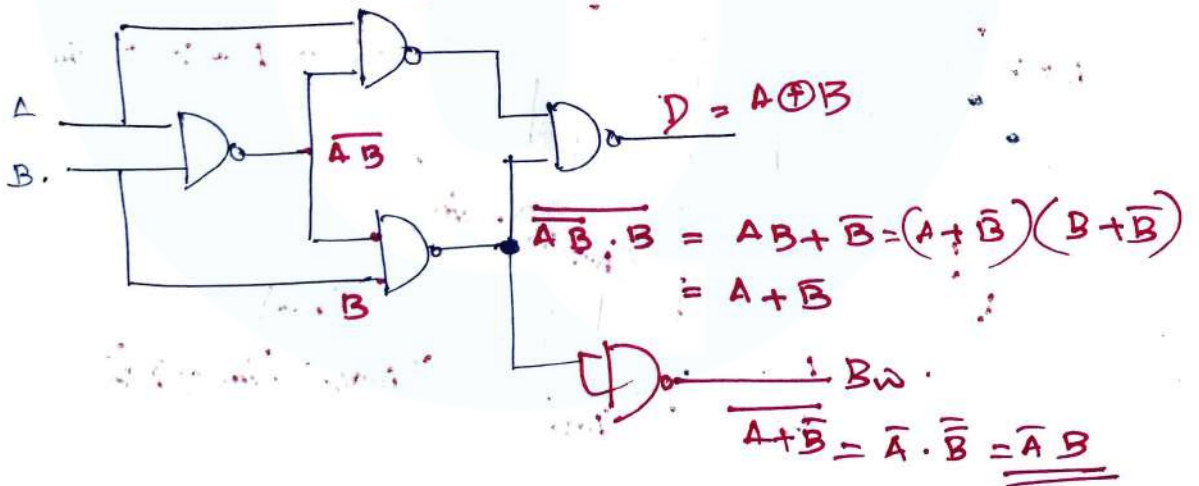
$$D = \bar{A}B + A\bar{B} = A \oplus B$$

$$Bw = \bar{A}B$$

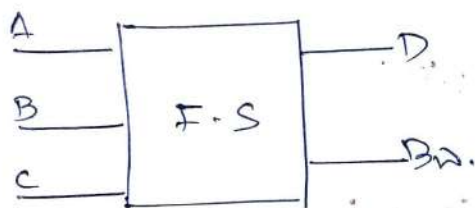
A	B	D	Bw
0	0	0	0
0	1	1 ✓	1 ✓
1	0	1 ✓	0
1	1	0	0



using NAND gate



Full Subtractor



	A	B	C	D	Bn
0	0	0	0	0	0
1	0	0	1	1	1
2	0	1	0	1	1
3	0	1	1	0	1
4	1	0	0	1	0
5	1	0	1	0	0
6	1	1	0	0	0
7	1	1	1	1	1

Difference

AB \ C	0	1
$\bar{A}\bar{B}$	0	1
$\bar{A}B$	1	0
AB	0	1
$A\bar{B}$	1	0

$$D = A \oplus B \oplus C \quad \checkmark$$

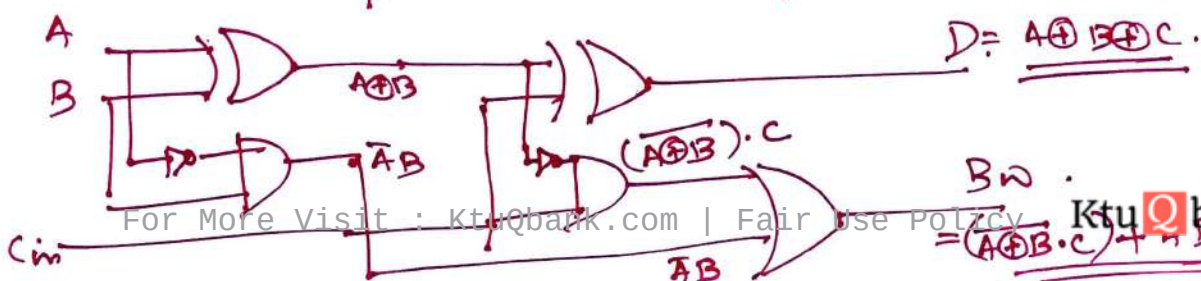
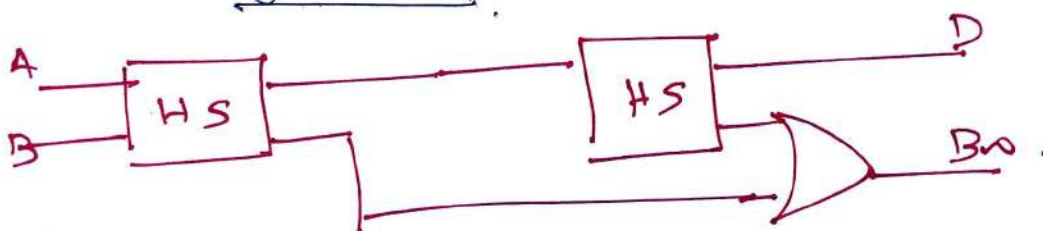
Borrow

AB \ C	0	1
$\bar{A}\bar{B}$	0	1
$\bar{A}B$	1	0
AB	0	1
$A\bar{B}$	1	0

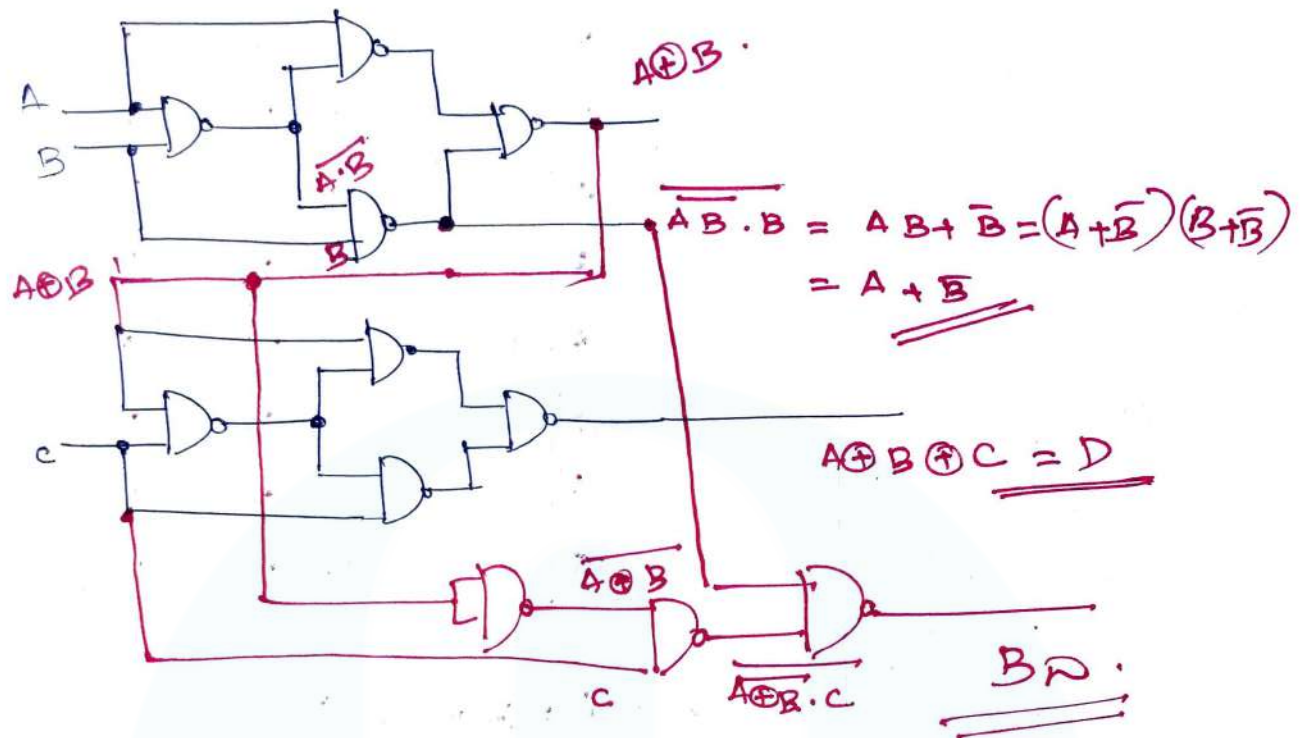
$$Bn = \bar{A}B + \bar{A}C + BC$$

$$\begin{aligned}
 Bn &= \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC + AB\bar{C} \\
 &= (\bar{A}\bar{B} + \bar{A}B)C + \bar{A}B(\bar{C} + C) \\
 &= \bar{A}(\bar{B} + B)C + \bar{A}B \\
 &= \bar{A}C + \bar{A}B \quad \checkmark
 \end{aligned}$$

Block diagram representation



Using NAND gate



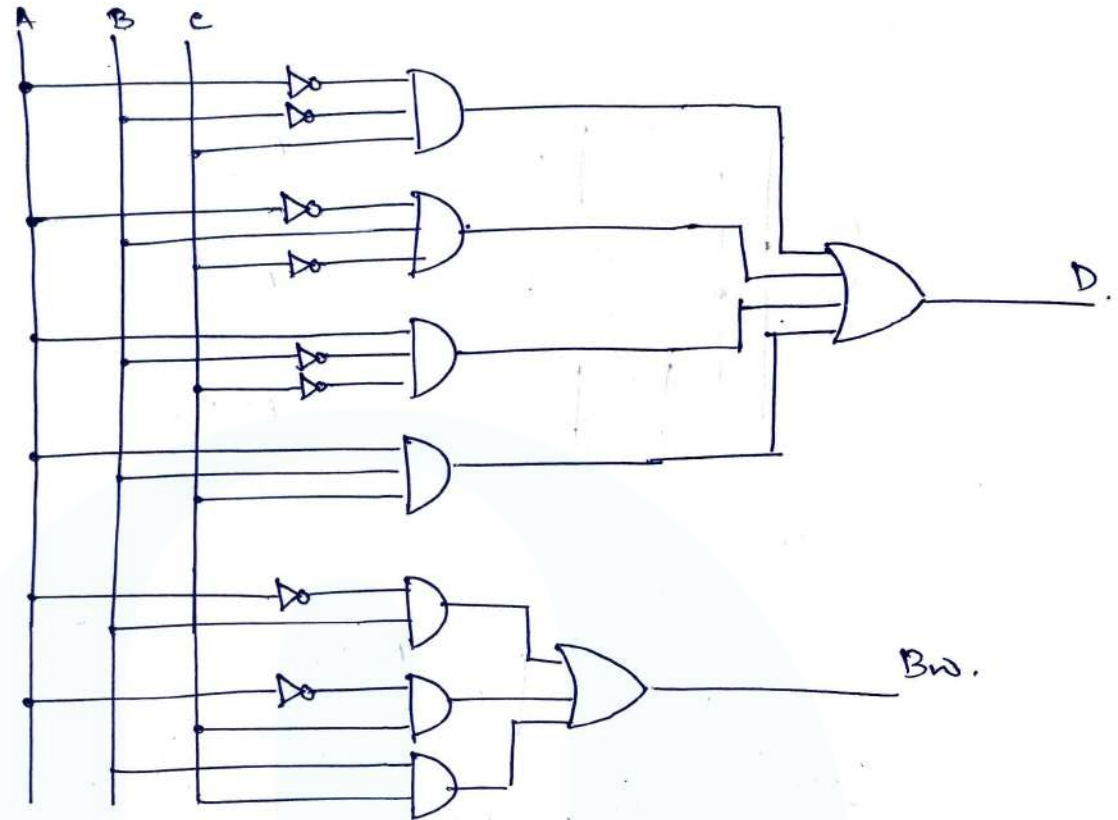
$$B_2 = \overline{(A \oplus B)} \cdot C + \overline{A} \cdot B$$

$$= \overline{A \oplus B} \cdot C + \overline{A} \cdot B$$

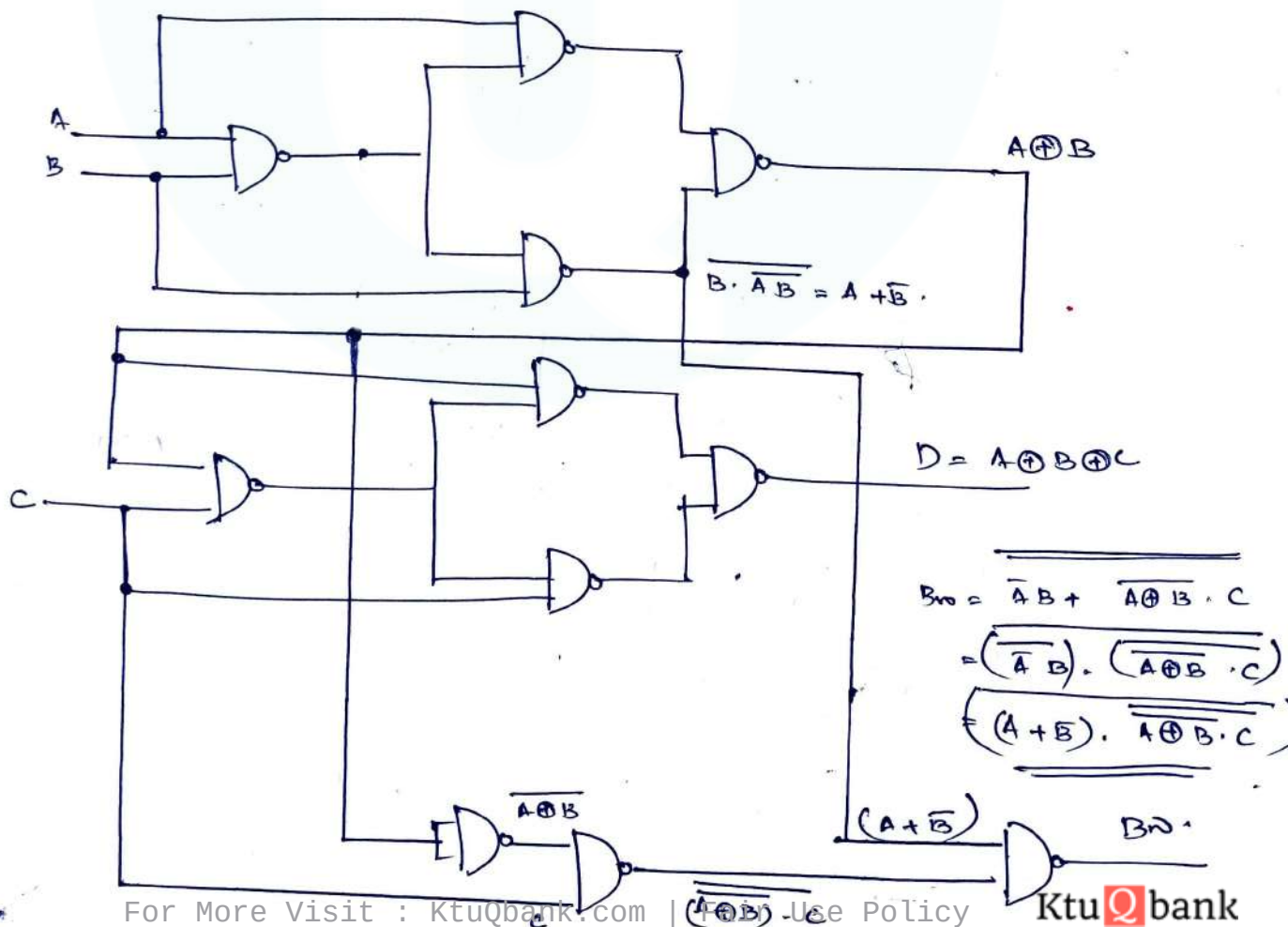
$$= \overline{A \oplus B} \cdot C \cdot (\overline{A} \cdot B)$$

$$= (\overline{A \oplus B} \cdot C) \cdot (A + B)$$

Full Subtractor using basic gates



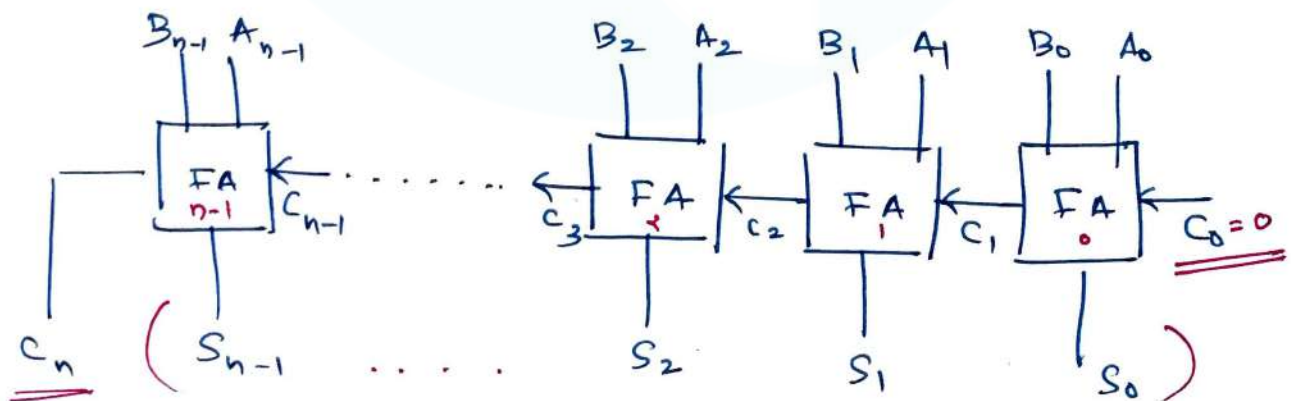
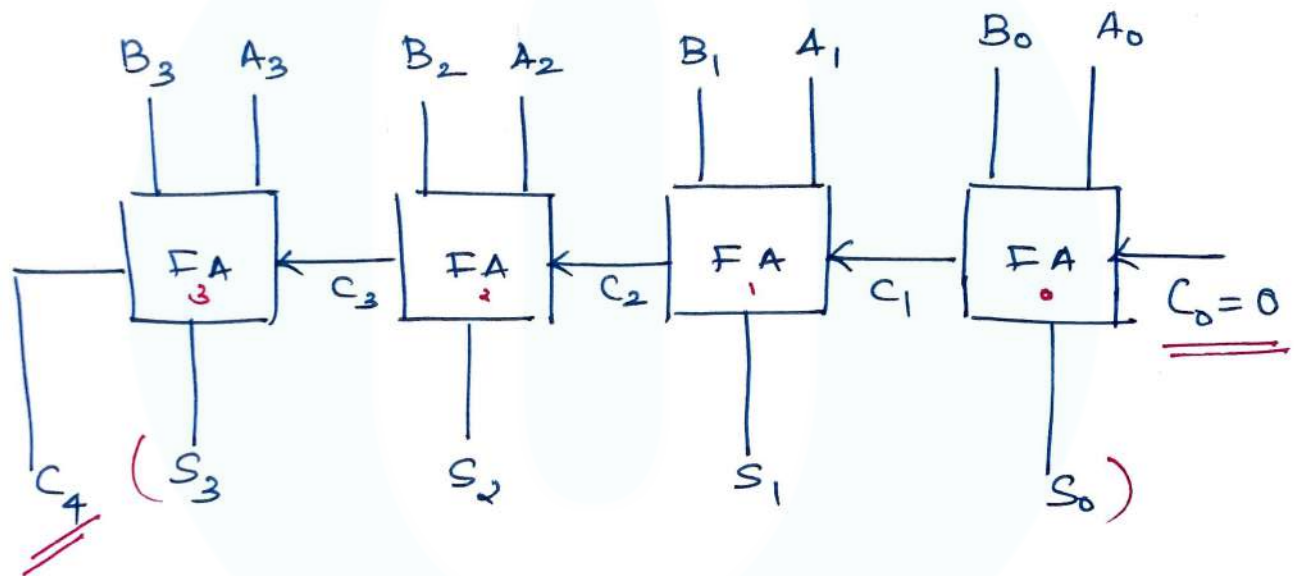
Full Subtractor using NAND gates



4 bit Parallel Adder

	C_3	C_2	C_1	$(C_0 = 0)$
	(1)	(1)	(1)	
$A \rightarrow$	A_3	A_2	A_1	A_0
	1	1	0	1
$B \rightarrow$	B_3	B_2	B_1	B_0
	0	1	1	1

C_4	S_3	S_2	S_1	S_0
1	0	1	0	0



BCD Adder

0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1

$$5 + 3 = \underline{8}$$

$$\begin{array}{r} 0101 + \\ 0011 \\ \hline 1000 = \underline{8} \end{array}$$

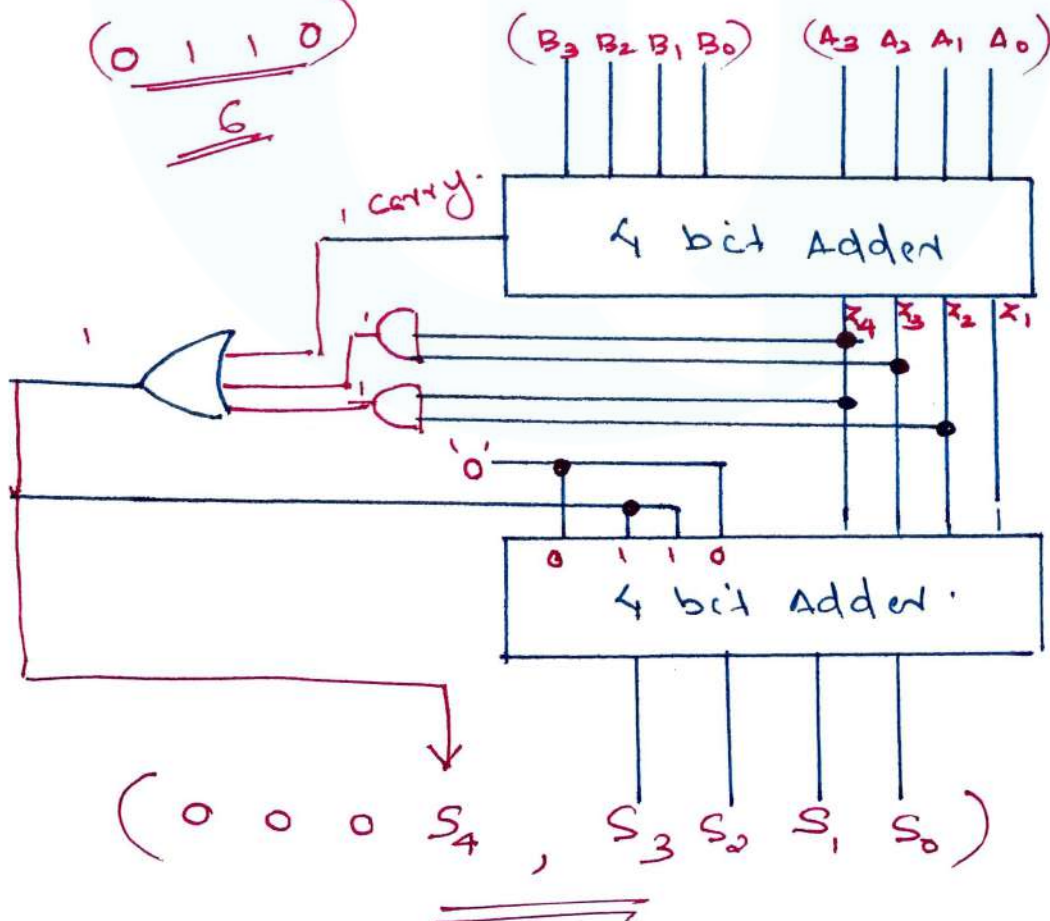
$$6 + 8 = \underline{14}$$

$$\begin{array}{r} 0110 + \\ 1000 \\ \hline 1110 \end{array} \quad \begin{array}{r} (1)(1) \\ 1110 \\ 0110 \\ \hline \end{array}$$

$$\begin{array}{r} 1001 + \\ 0110 \\ \hline 10000 + \\ 0110 \\ \hline \end{array} \quad \begin{array}{r} (1)(1)(1) \\ 1001 \\ \hline \end{array} \quad \begin{array}{r} (1)(1) \\ 10000 + \\ 0110 \\ \hline \end{array}$$

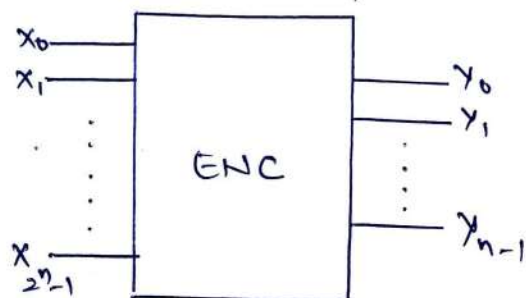
$$(9+7) = \underline{16}$$

$$\underline{(0110)} \\ \underline{6}$$



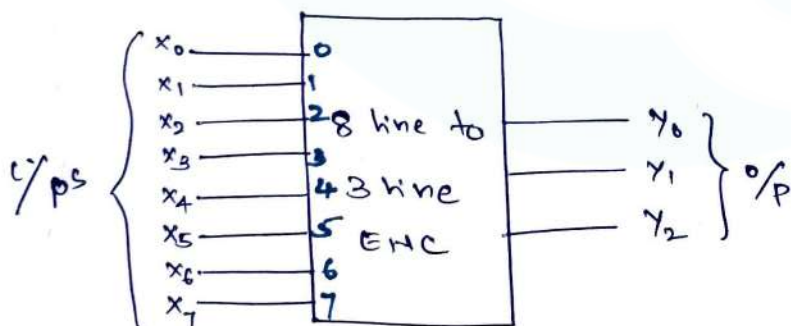
ENCODERS.

Encoder is a device whose i/p's are decimal digits or alphabetic characters and whose o/p's are coded representation of these i/p's. i.e, it is a device which converts the familiar number or symbols into coded formats.



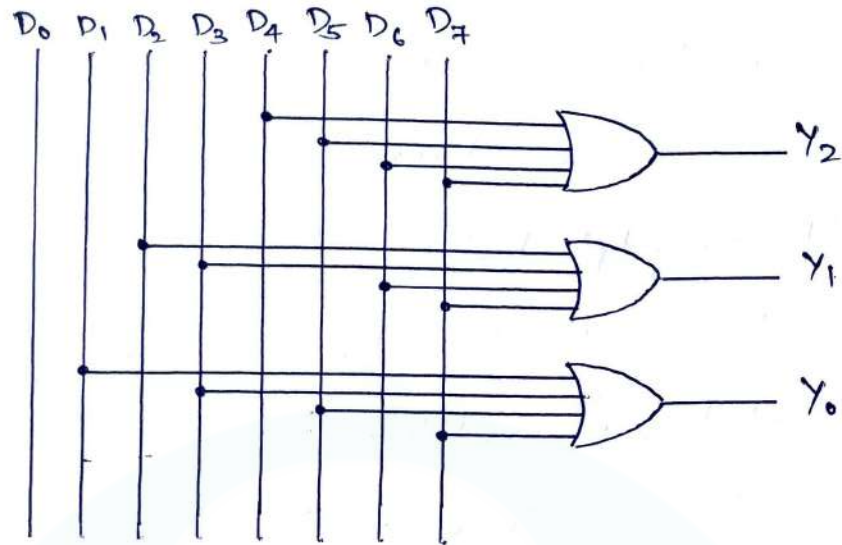
An encoder has a no. of i/p lines, only one of which is activated at a given time and produces an n-bit o/p code depending on which i/p is activated.

8 Line to 3 line ENC

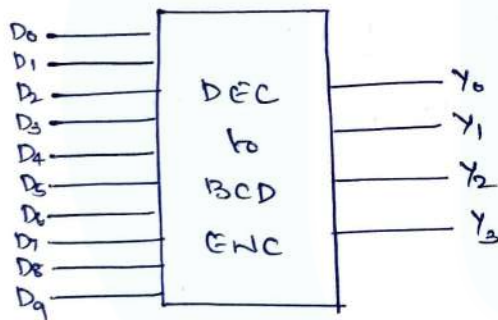


i/p	o/p		
	y_2	y_1	y_0
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1

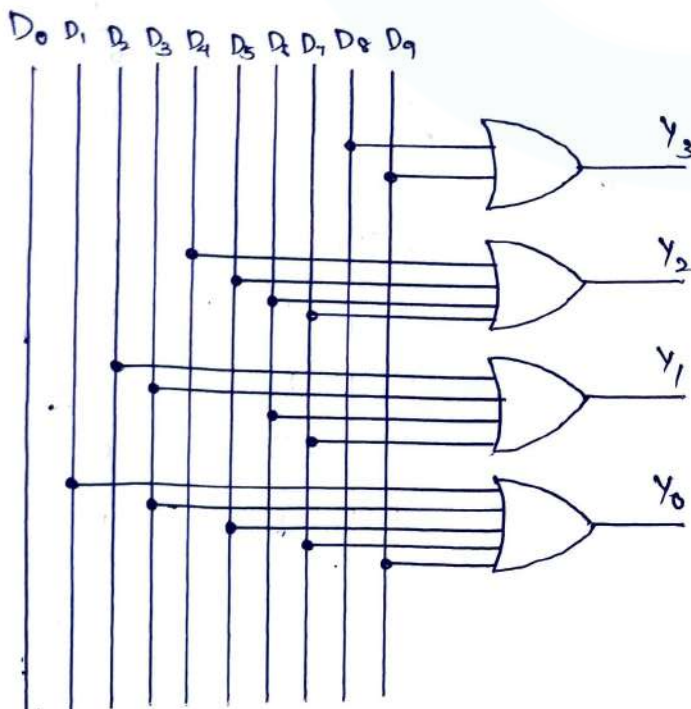
Logic Diagram of 8 line to 3 line.



Decimal to BCD Encoder.

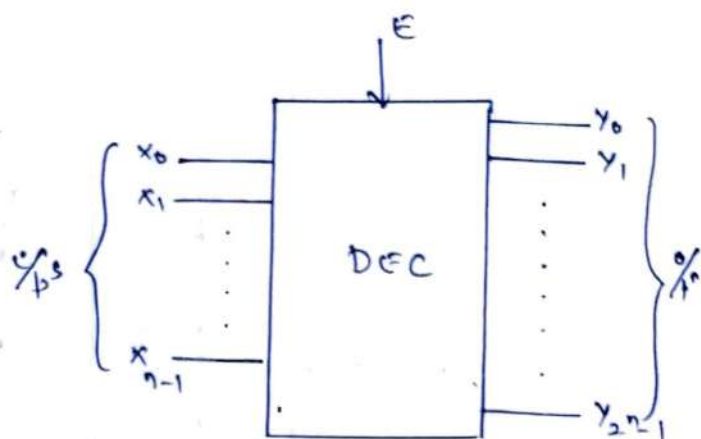


$\%D$	$\%P$			
	Y_3	Y_2	Y_1	Y_0
D_0	0	0	0	0
D_1	0	0	0	1
D_2	0	0	1	0
D_3	0	0	1	1
D_4	0	1	0	0
D_5	0	1	0	1
D_6	0	1	1	0
D_7	0	1	1	1
D_8	1	0	0	0
D_9	1	0	0	1



DECODERS.

A decoder is a circuit that converts an n -bit binary code into M -o/p lines. The max. value of M is 2^n . The decoder identifies

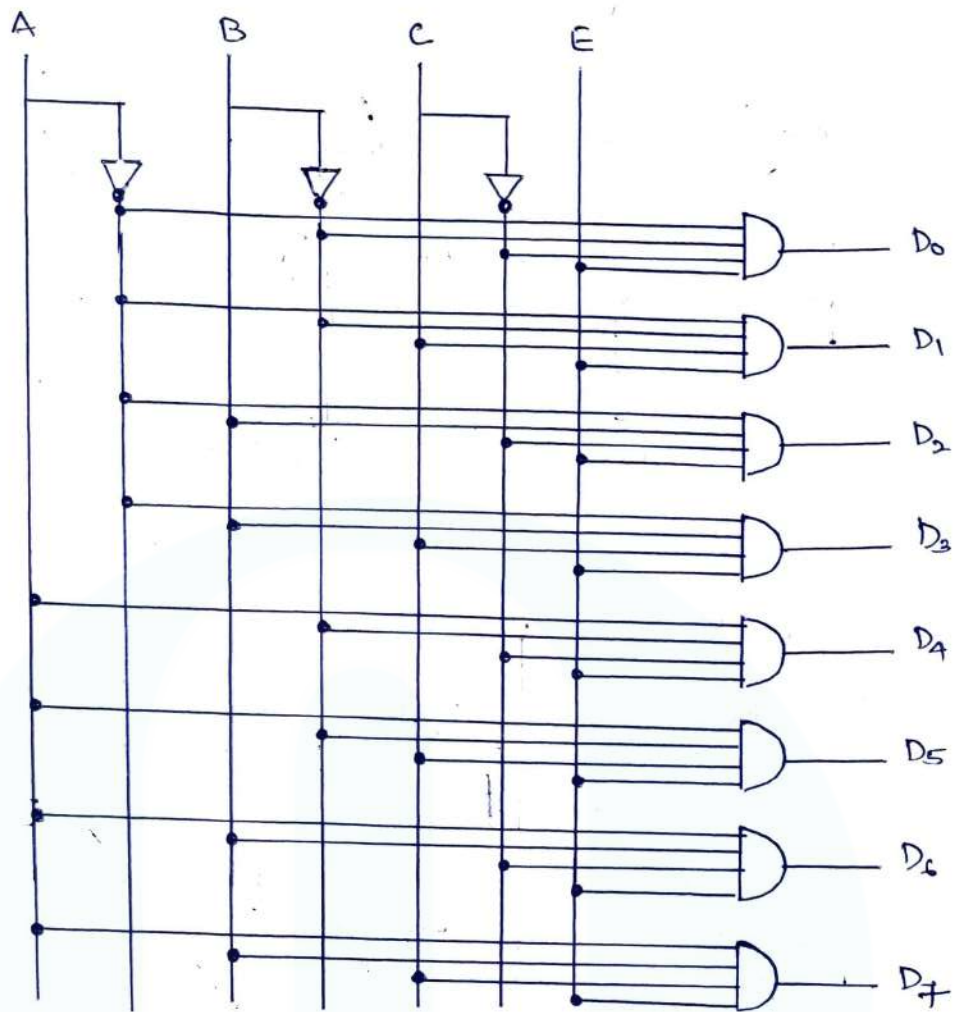


or recognize or detects a particular code. There are 2^n possible combinations of i/p. For each combination only one of the M o/p's will be active (high) and all other o/p's will remain inactive (low).

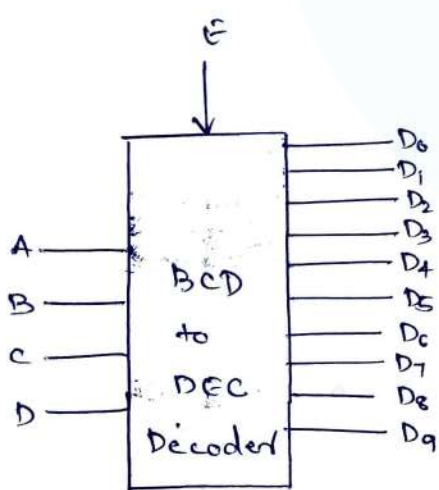
3 Line to 8 Line Decoder.

i/p			o/p							
A	B	C	D ₀	D ₁	D ₂	D ₃	D ₄	D ₅	D ₆	D ₇
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1

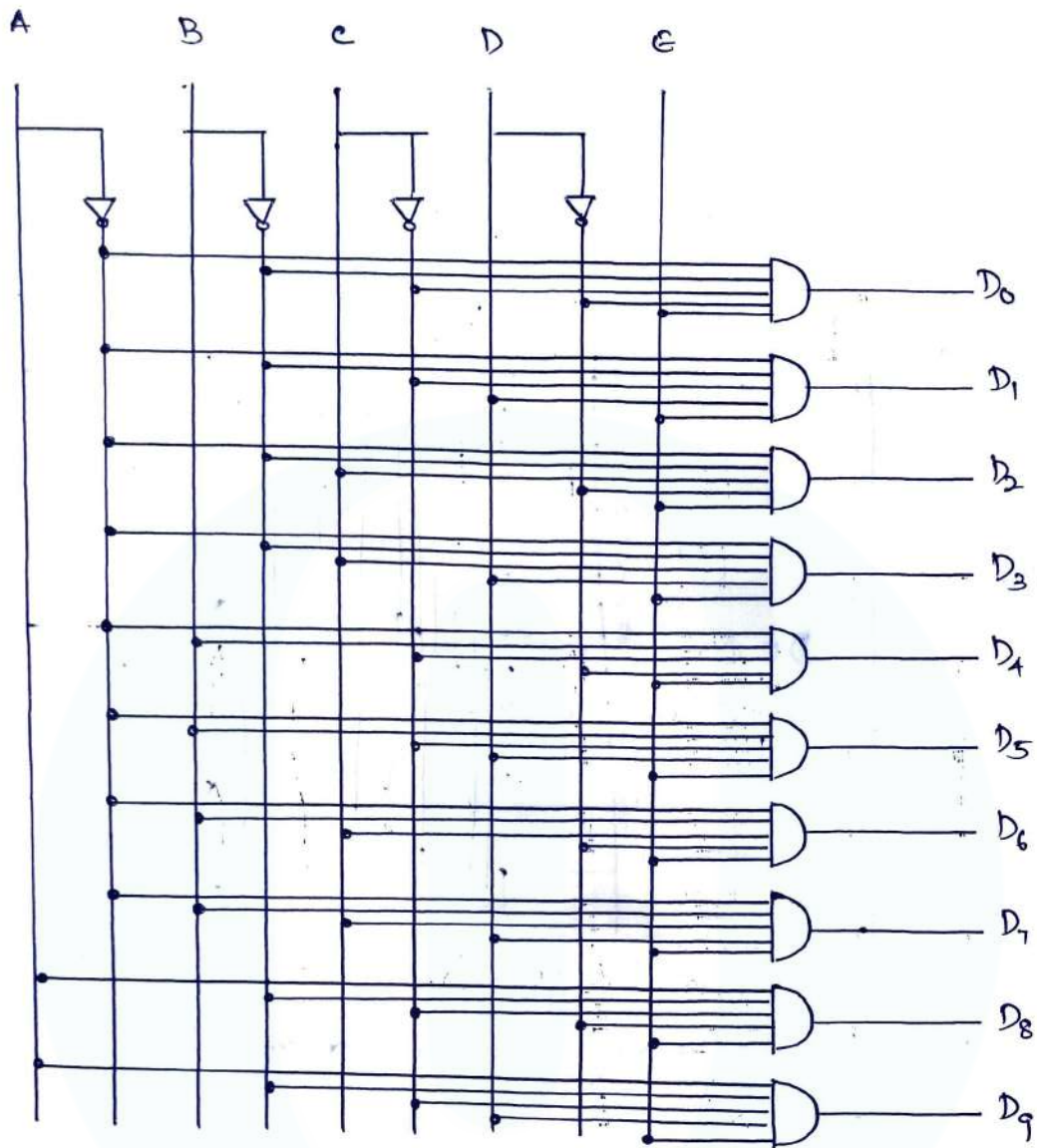
The decoder have a special i/p known as enable i/p. The decoder will active if, $E = 1$ and will inactive if $E = 0$.



BCD to Decimal Decoder



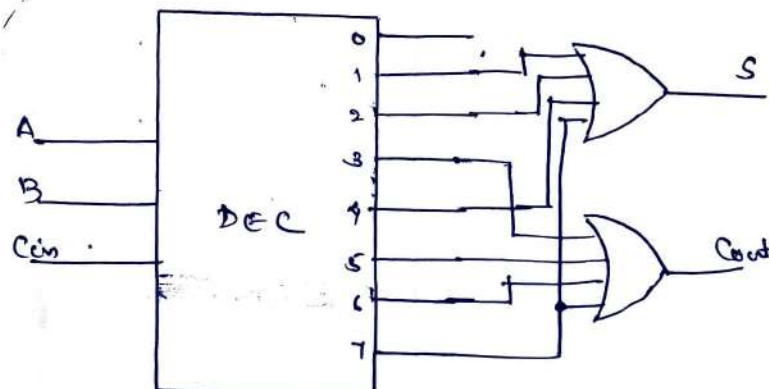
i/p				o/p									
A	B	C	D	D ₀	D ₁	D ₂	D ₃	D ₄	D ₅	D ₆	D ₇	D ₈	D ₉
0	0	0	0	1	0	0	0	0	0	0	0	0	0
0	0	0	1	0	1	0	0	0	0	0	0	0	0
0	0	1	0	0	0	1	0	0	0	0	0	0	0
0	0	1	1	0	0	0	1	0	0	0	0	0	0
0	1	0	0	0	0	0	0	1	0	0	0	0	0
0	1	0	1	0	0	0	0	0	1	0	0	0	0
0	1	1	0	0	0	0	0	0	0	1	0	0	0
0	1	1	1	0	0	0	0	0	0	0	1	0	0
1	0	0	0	0	0	0	0	0	0	0	0	1	0
1	0	0	1	0	0	0	0	0	0	0	0	0	1

Ckt DiagramImplementation of Logic functions using Decoders.

Q:- Full Adder.

$$S = \sum m(1, 2, 4, 7)$$

$$Cout = \sum m(3, 5, 6, 7)$$

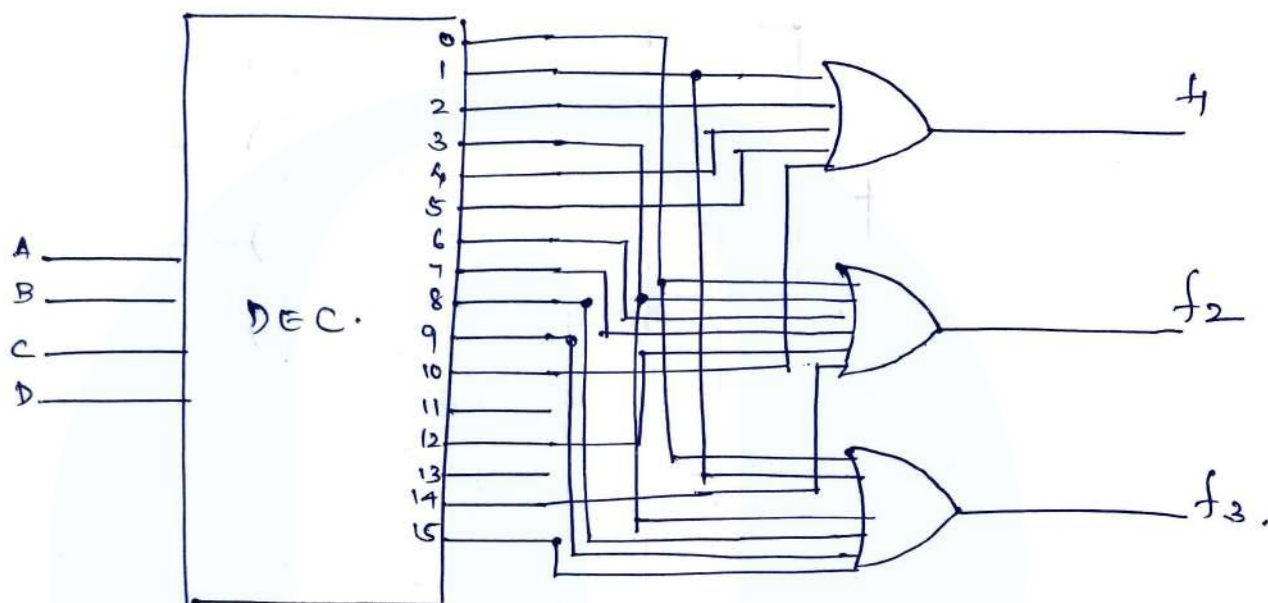


Q:- Implement the boolean expression on one decoder.

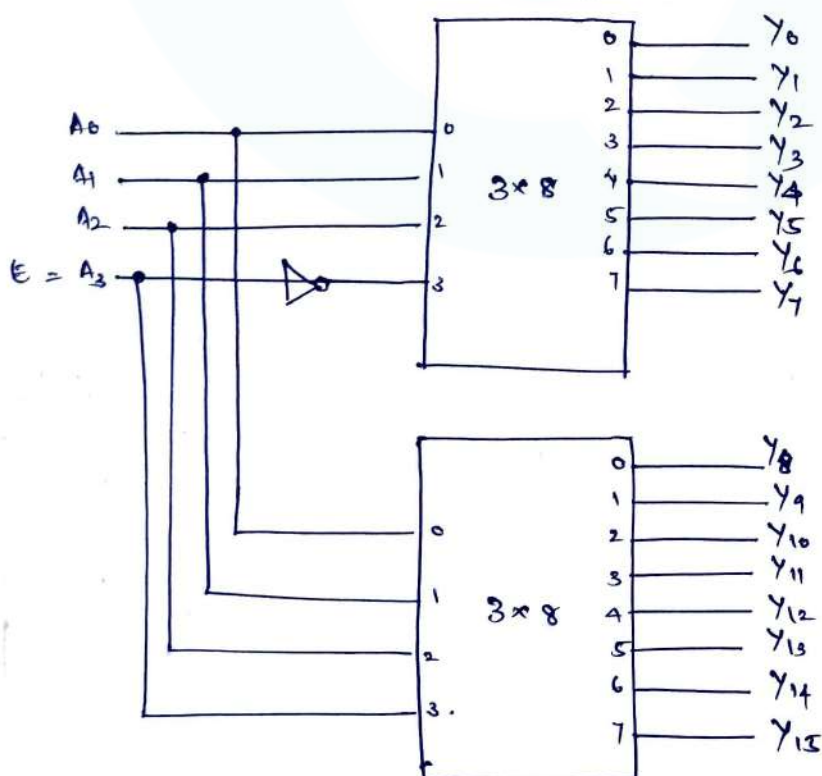
$$f_1 = \sum m(1, 2, 4, 5, 10)$$

$$f_2 = \sum m(0, 3, 6, 7, 12, 14)$$

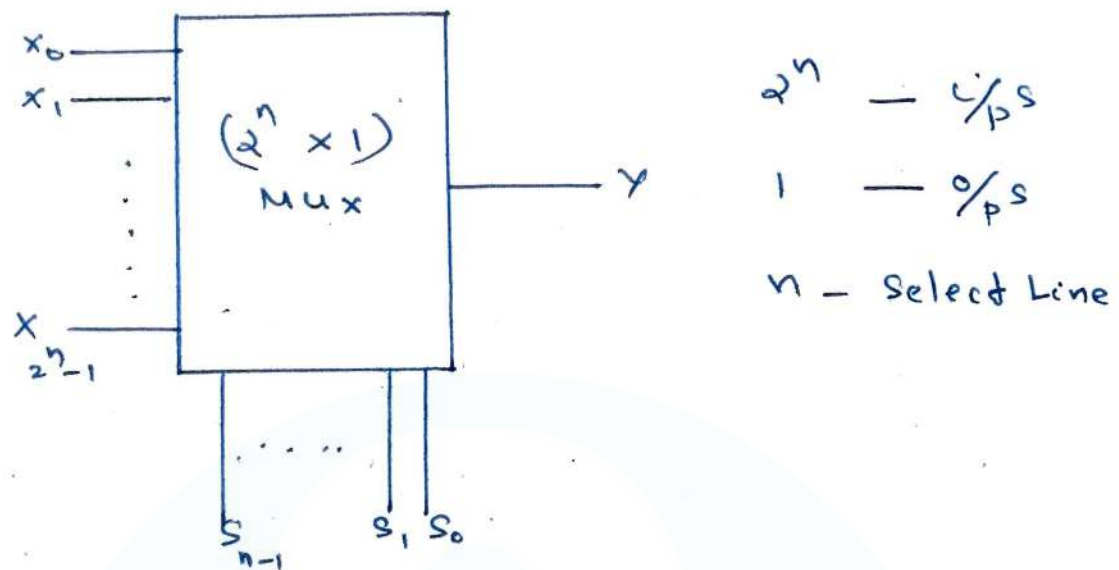
$$f_3 = \sum m(0, 1, 3, 8, 9, 15)$$



Q:- 4 Line to 16 Line decoder from 3 line 8 line decoder.

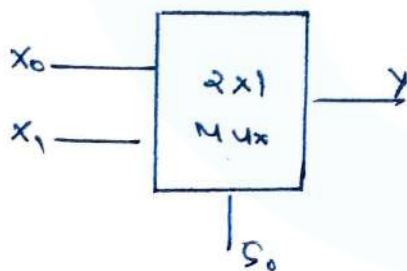


MULTIPLEXER / SELECTOR

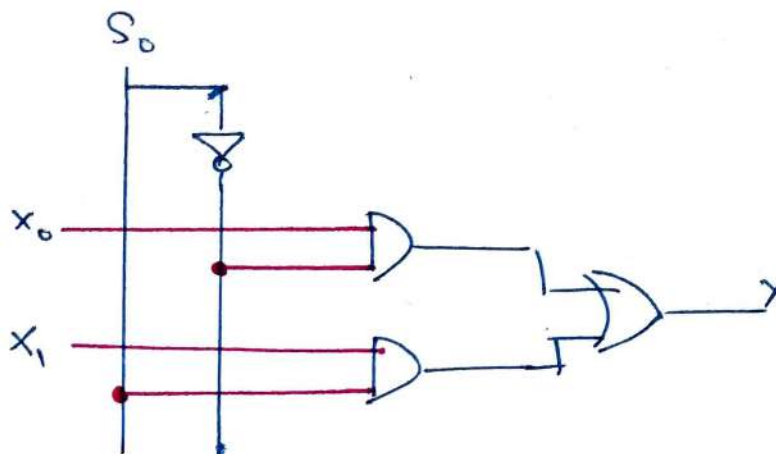


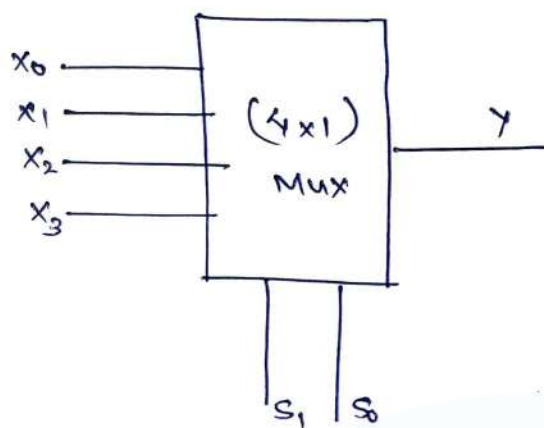
Mux is 'many to One' converted. Select lines select i/p to o/p. So it is called selector.

(2x1) Mux.

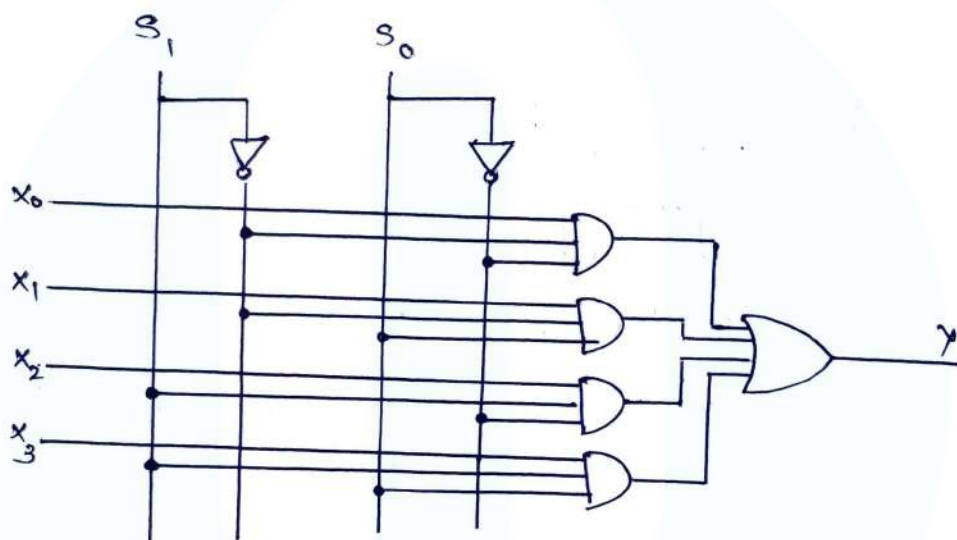
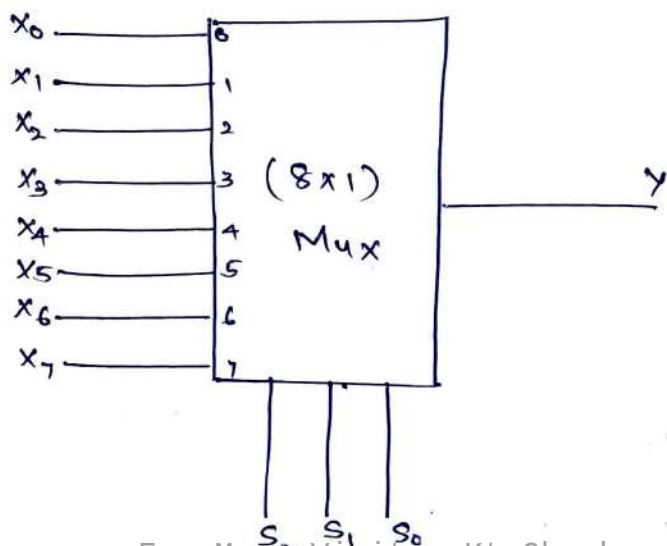


s_0	y
0	x_0
1	x_1

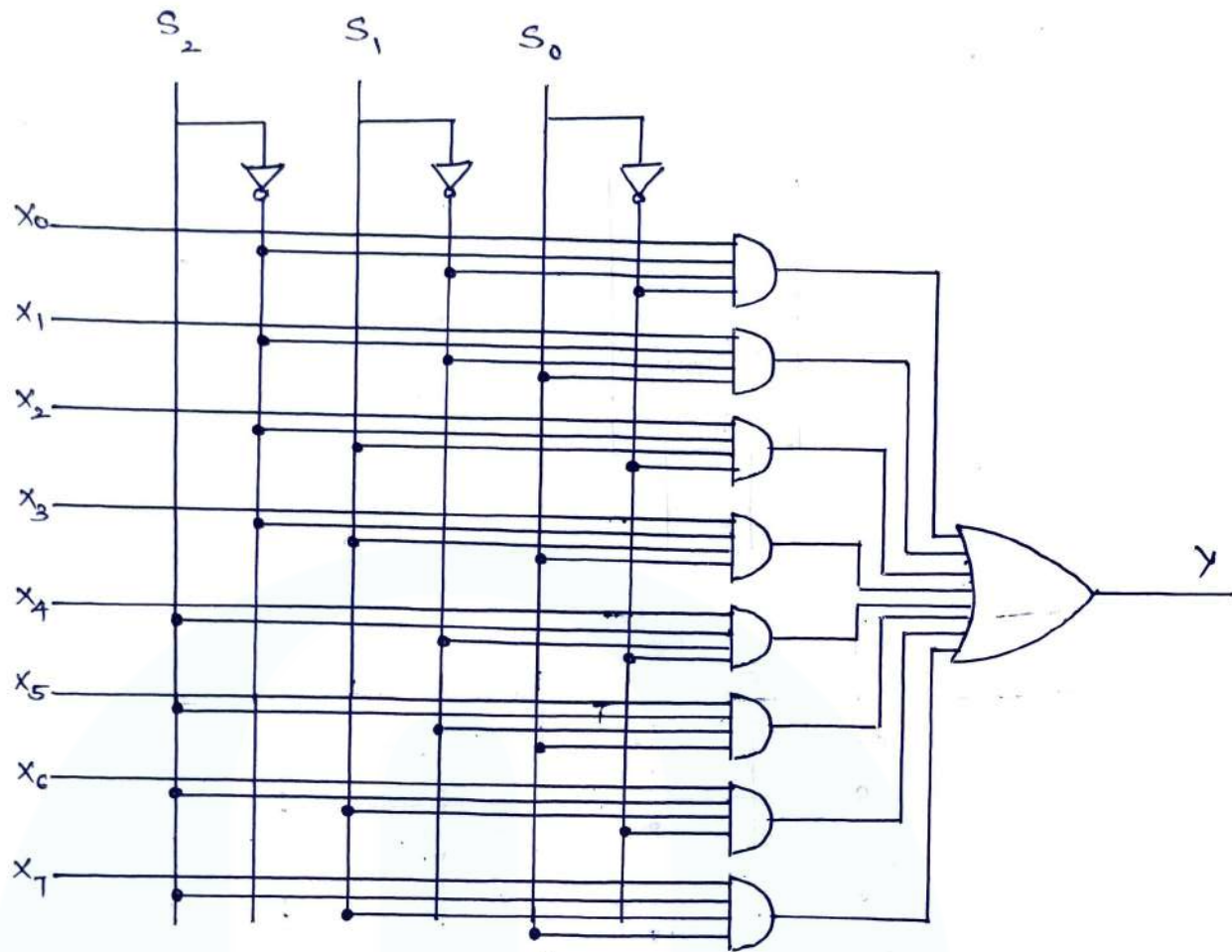


(4x1) Mux

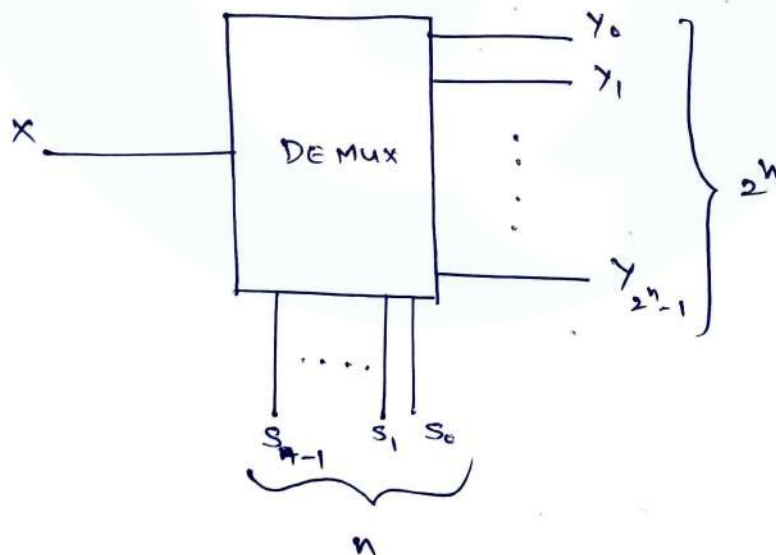
s_1	s_0	y
0	0	x_0
0	1	x_1
1	0	x_2
1	1	x_3

(8x1) Mux

s_2	s_1	s_0	y
0	0	0	x_0
0	0	1	x_1
0	1	0	x_2
0	1	1	x_3
1	0	0	x_4
1	0	1	x_5
1	1	0	x_6
1	1	1	x_7

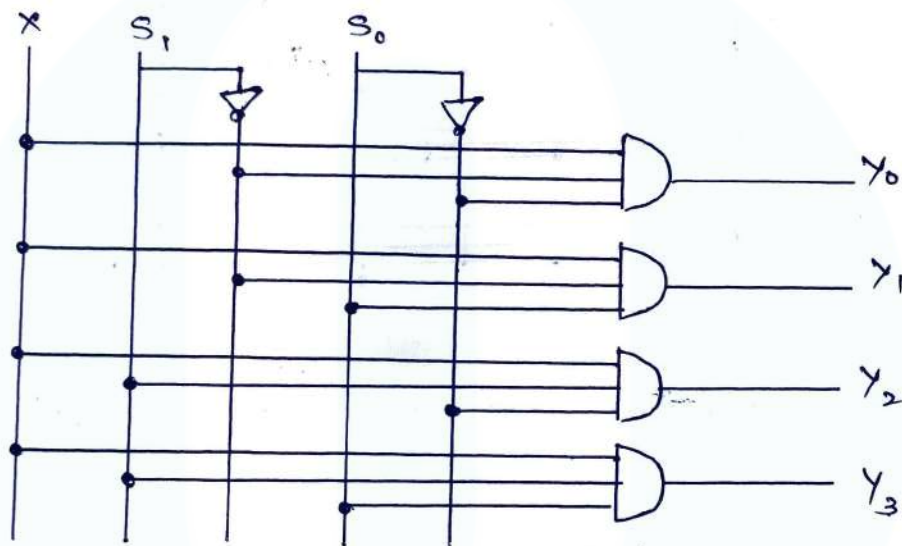
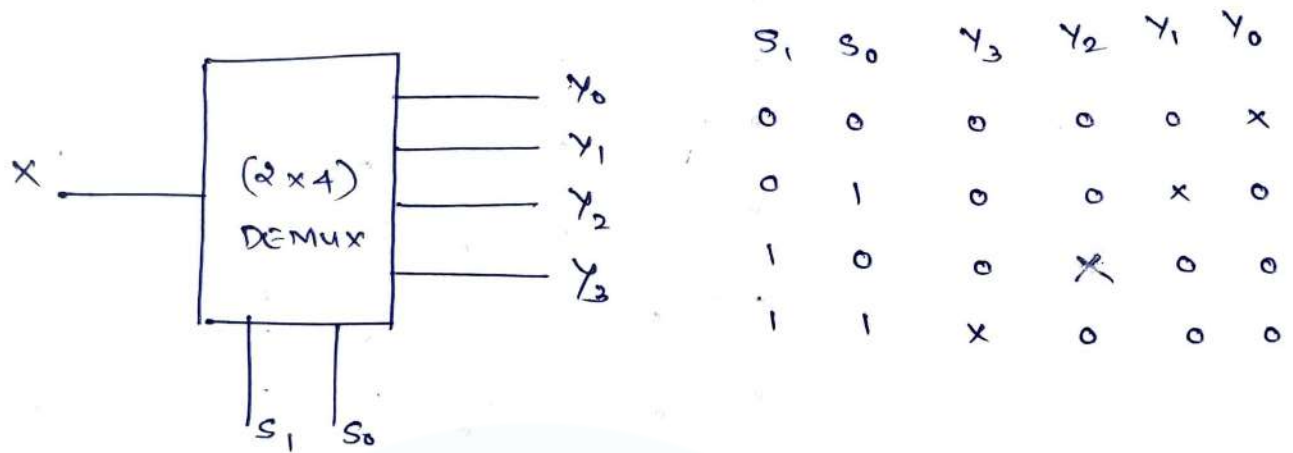


DEMULTIPLEXER / DISTRIBUTOR.

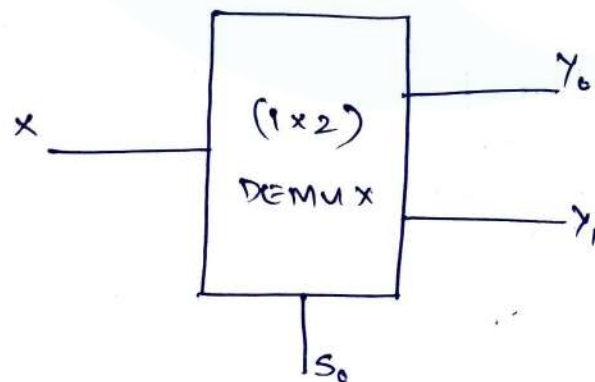


A demultiplexer has one i/p x , 2^n o/p $(y_0, y_1, \dots, y_{2^n-1})$ and n -select line $(s_0, s_1, \dots, s_{n-1})$.

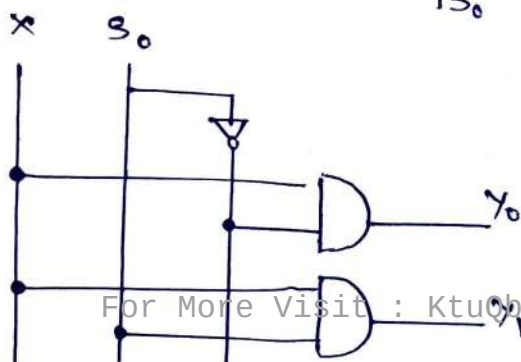
1 line to 4 line demultiplexer or (2×4) .



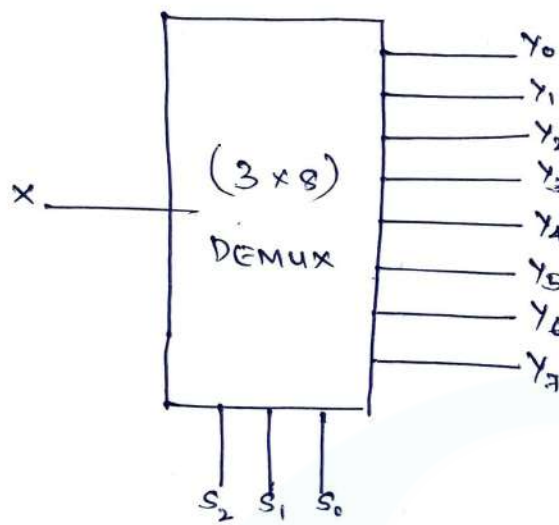
1 line to 2 line Demultiplexer (1×2) .



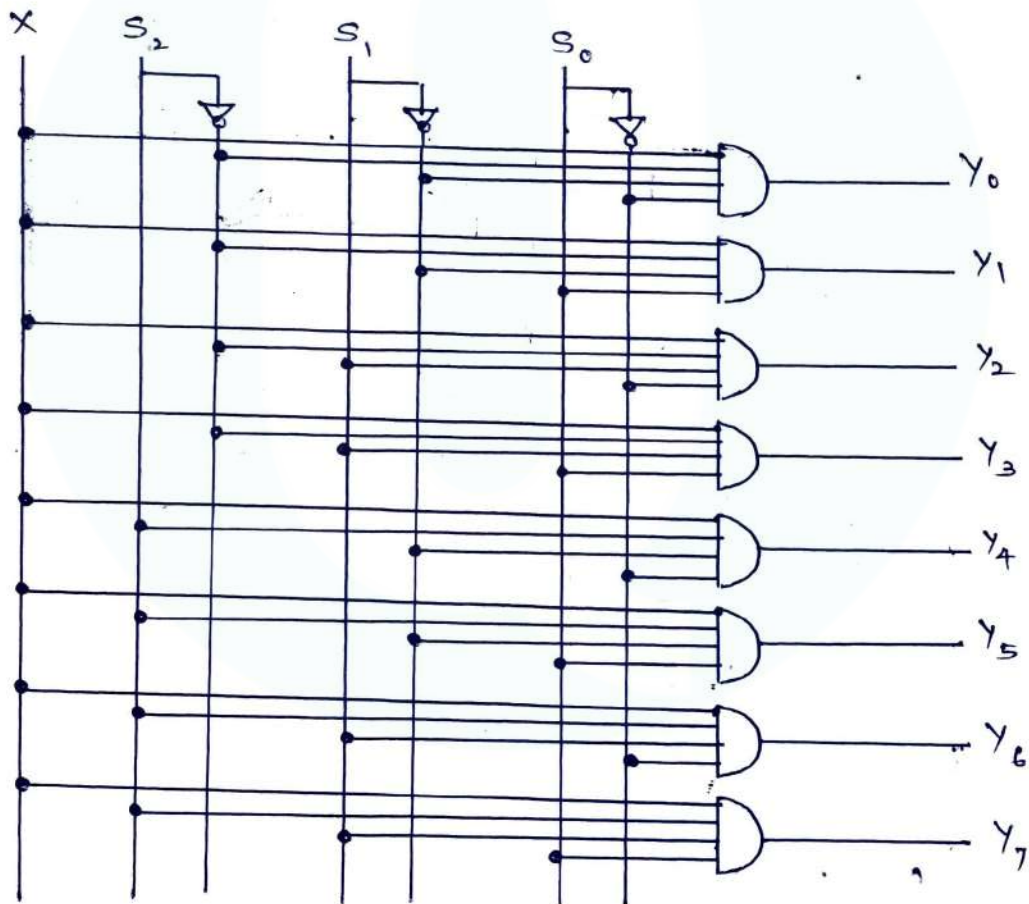
S_0	Y_1	Y_0
0	0	x
1	x	0



1 Line to 8 Line Demultiplexer (3x8).



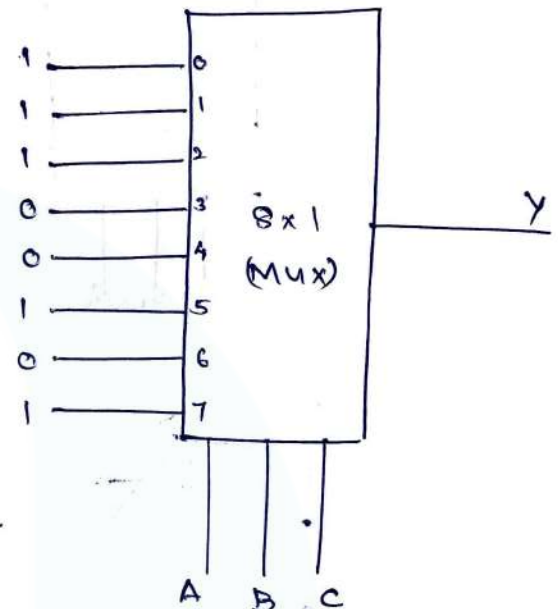
S_2	S_1	S_0	Y_7	Y_6	Y_5	Y_4	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	0	0	0	0	0	X
0	0	1	0	0	0	0	0	0	X	0
0	1	0	0	0	0	0	0	X	0	0
0	1	1	0	0	0	0	X	0	0	0
1	0	0	0	0	0	X	0	0	0	0
1	0	1	0	0	X	0	0	0	0	0
1	1	0	0	X	0	0	0	0	0	0
1	1	1	X	0	0	0	0	0	0	0



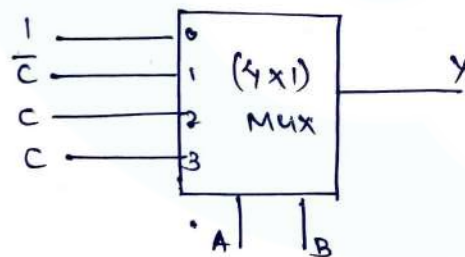
Implementation of Boolean Expressions using Mux

Q:- $Y = \sum m(0, 1, 2, 5, 7)$ using (8x1) Mux.

	A	B	C	Y
0	0	0	0	1
1	0	0	1	1
2	0	1	0	1
3	0	1	1	0
4	1	0	0	0
5	1	0	1	1
6	1	1	0	0
7	1	1	1	1

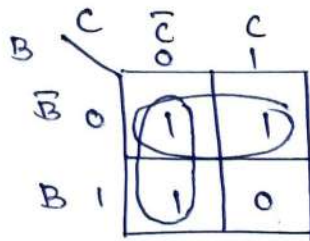


Q:- Implement $Y = \sum m(0, 1, 2, 5, 7)$ using (4x1) Mux.

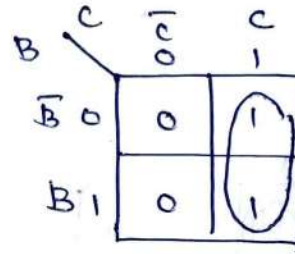


Q:- Implement $Y = \sum m(0, 1, 2, 5, 7)$ using (2x1) Mux.

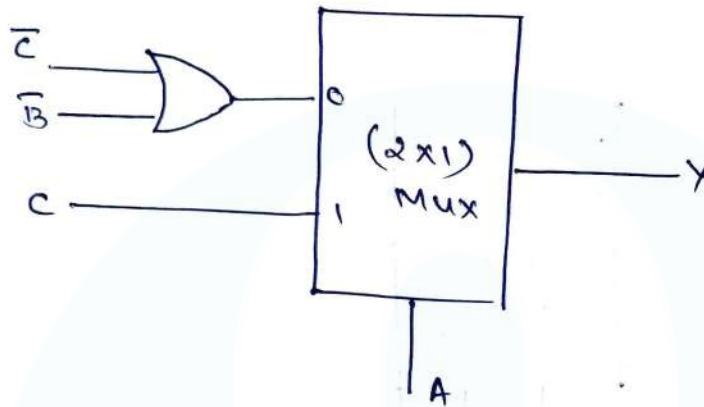
	A	B	C	Y
0	0	0	0	1
1	0	0	1	1
2	0	1	0	1
3	0	1	1	0
4	1	0	0	0
5	1	0	1	1
6	1	1	0	0
7	1	1	1	1



$$Y = \bar{C} + \bar{B}$$

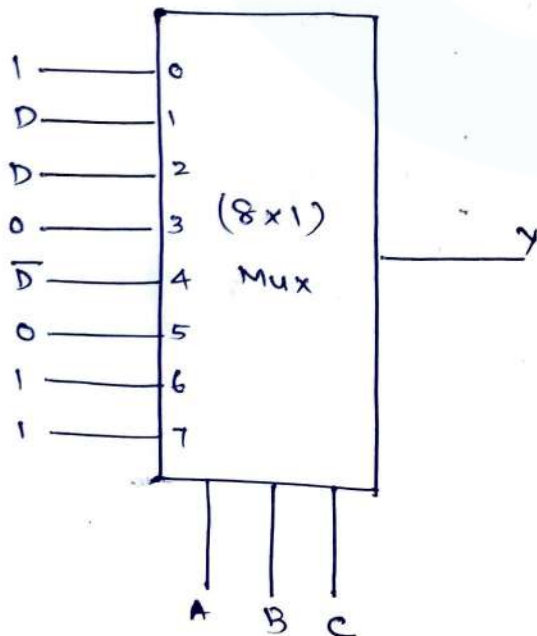


$$Y = C$$



Q.- Implement $Y = \sum m(0, 1, 3, 5, 8, 12, 13, 14, 15)$

Using (8x1) Mux.



	A	B	C	D	Y	
0	0	0	0	0	1	
1	0	0	0	1	1	1
2	0	0	1	0	0	
3	0	0	1	1	1	D
4	0	1	0	0	0	
5	0	1	0	1	1	D
6	0	1	1	0	0	
7	0	1	1	1	0	0
8	1	0	0	0	1	
9	1	0	0	1	0	D-bar
10	1	0	1	0	0	
11	1	0	1	1	0	0
12	1	1	0	0	1	
13	1	1	0	1	1	1
14	1	1	1	0	1	
15	1	1	1	1	1	1

Using (4x1) Mux.

$\bar{D}$	$\bar{D}$	D
$\bar{C}$	1	1
C	0	1

$$Y = D + \bar{C}$$

$\bar{D}$	$\bar{D}$	D
$\bar{C}$	0	1
C	0	0

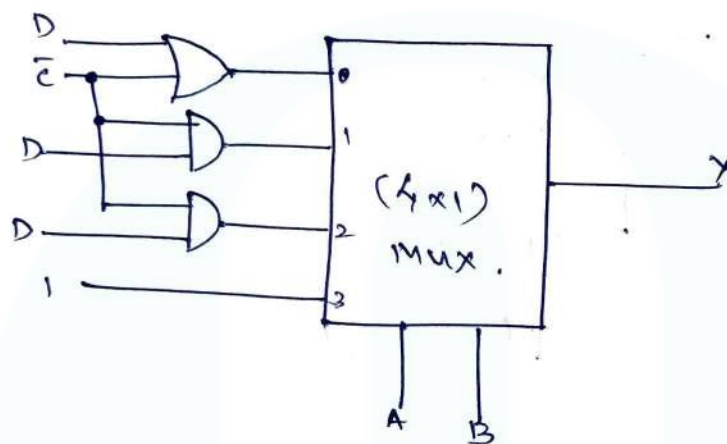
$$Y = \bar{C} D$$

$\bar{D}$	$\bar{D}$	D
$\bar{C}$	1	0
C	0	0

$$Y = \bar{C} \bar{D}$$

$\bar{D}$	$\bar{D}$	D
$\bar{C}$	1	1
C	1	1

$$Y = 1$$



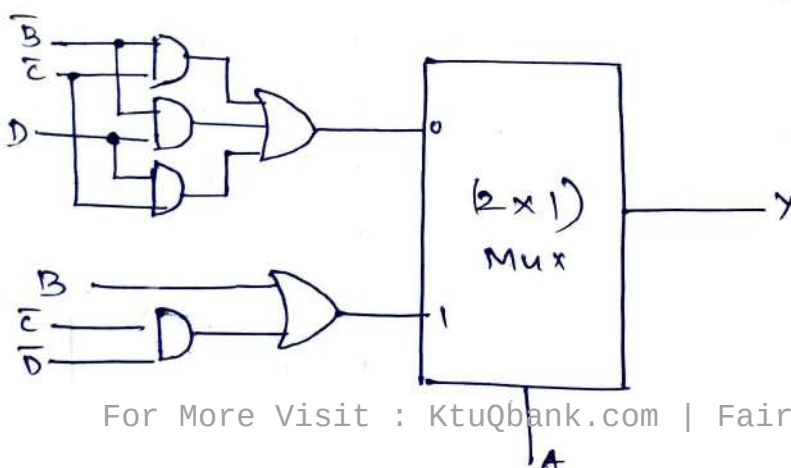
(2x1) Mux.

$\bar{D}$	$\bar{D}$	D
$\bar{B}\bar{C}$	0	1
$\bar{B}C$	2	0
BC	6	0
$B\bar{C}$	4	0

$$Y = \bar{B}\bar{C} + \bar{B}D + \bar{C}D$$

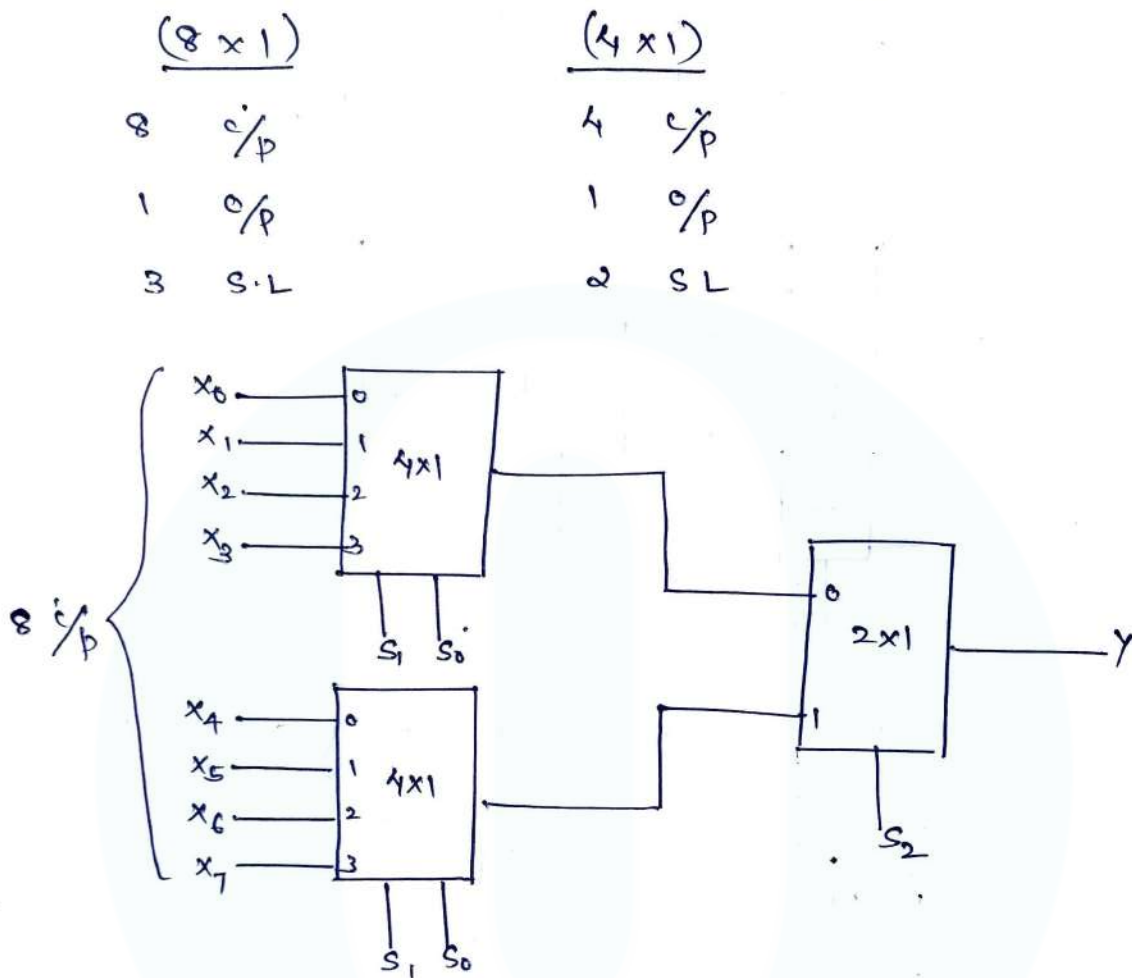
$\bar{D}$	$\bar{D}$	D
$\bar{B}\bar{C}$	0	1
$\bar{B}C$	2	0
BC	6	1
$B\bar{C}$	4	1

$$Y = B + \bar{C}\bar{D}$$

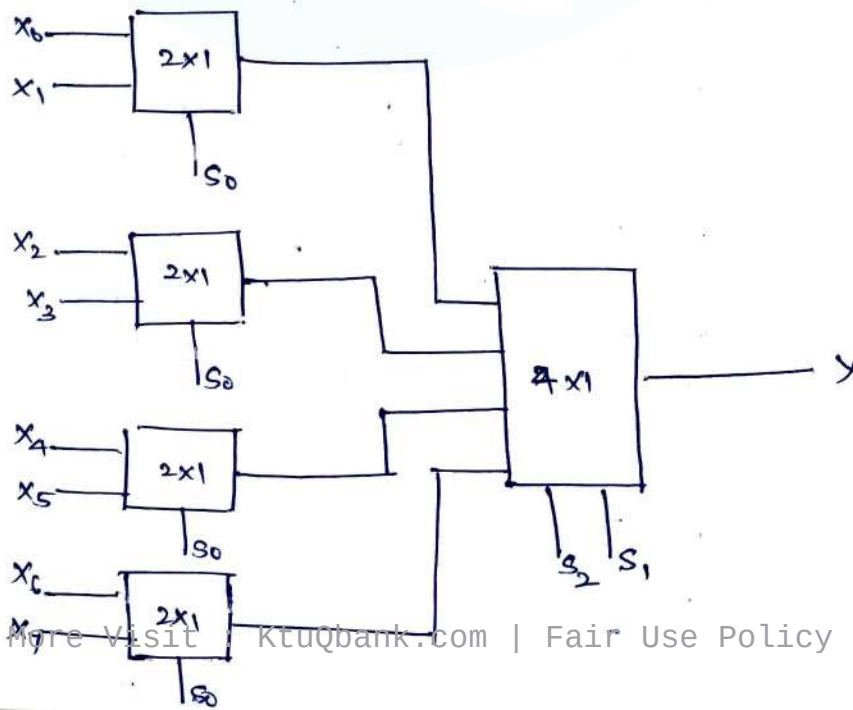


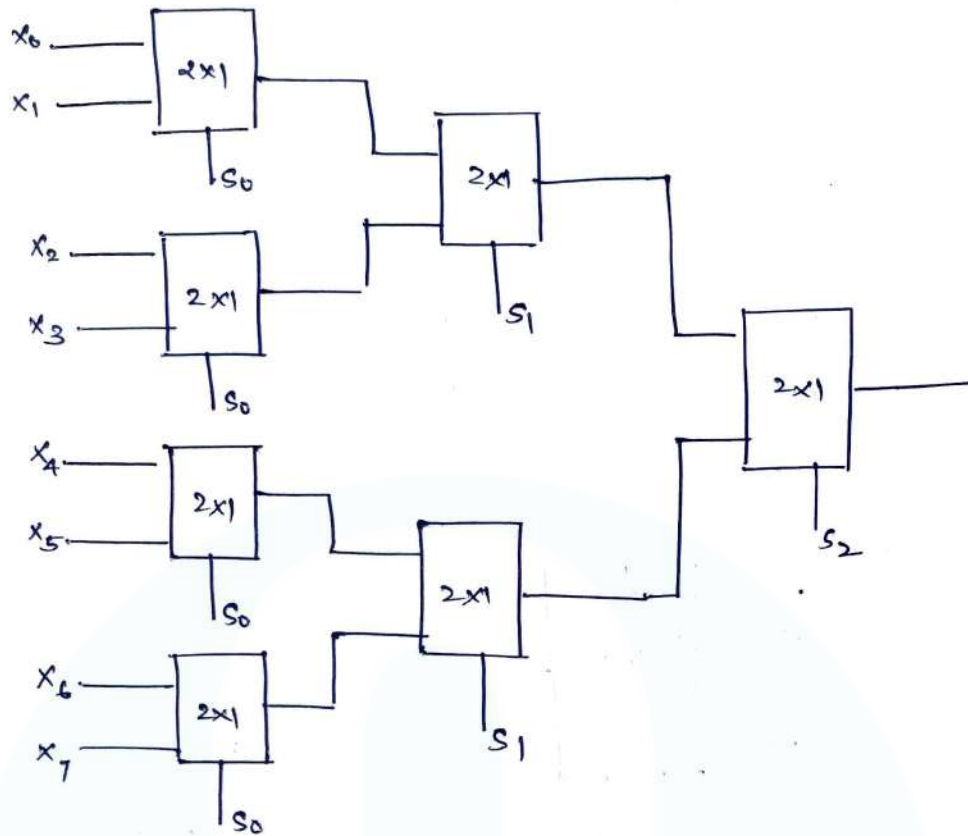
Mux Expansion.

Q:- Implement a (8×1) Mux using (4×1) Mux.

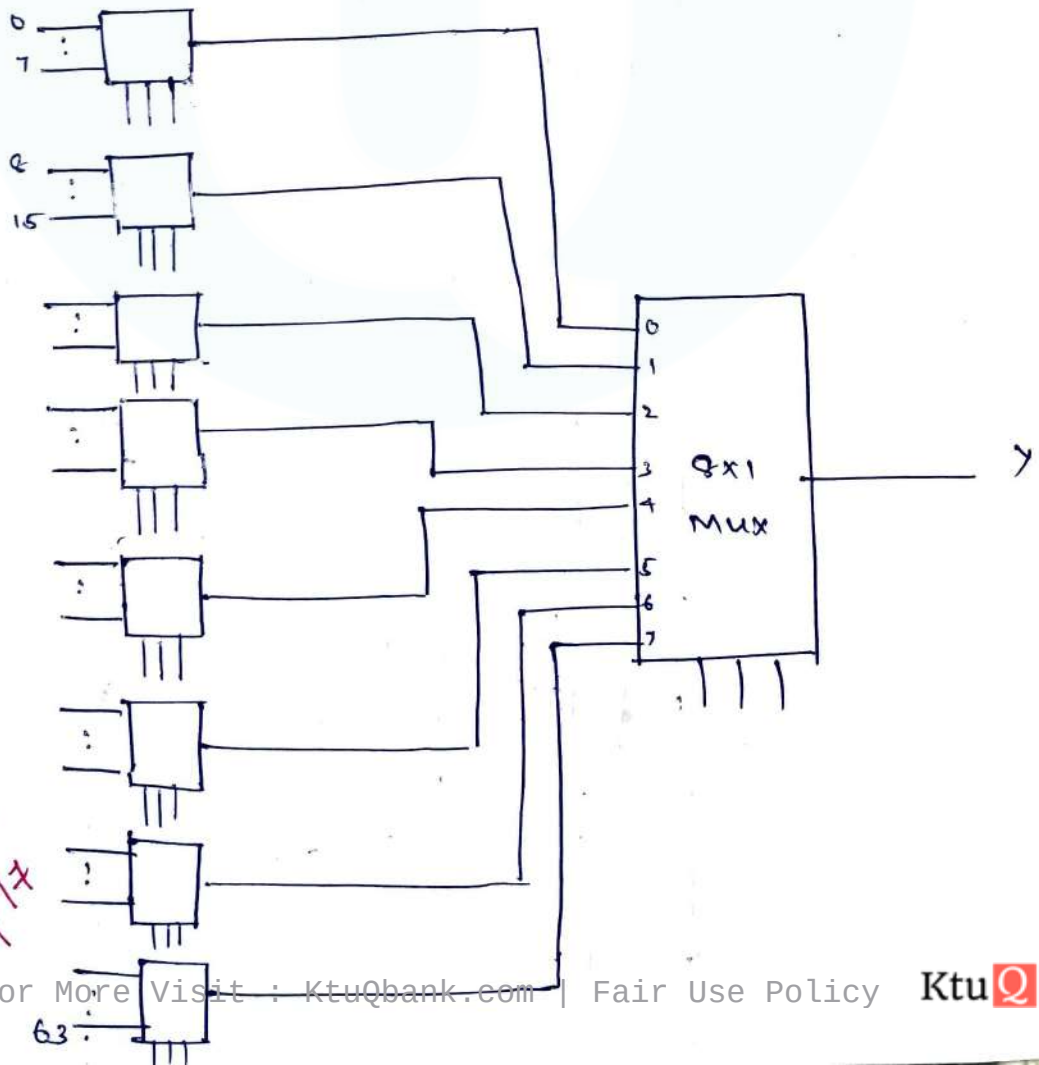


Q:- Implement a (8×1) Mux using (2×1) Mux & (4×1) Mux





Implement a (64×1) Mux using (8×1) Mux.



3/18/17

COMPARATORS.

1 Bit

$A_0 \cdot B_0$	G	L	E
0 0	0	0	1
0 1	0	1	0
1 0	1	0	0
1 1	0	0	1

$A \rightarrow A_0$

$B \rightarrow B_0$

If $A > B$ $G=1$

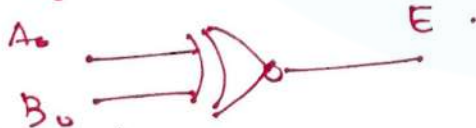
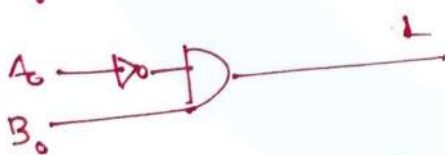
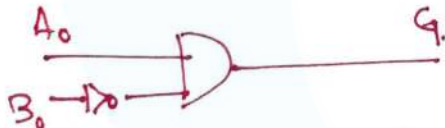
If $A < B$ $L=1$

If $A = B$ $E=1$

$$\underline{G} = A_0 \cdot \overline{B_0}$$

$$L = \overline{A_0} B_0$$

$$E = \overline{A_0 \oplus B_0} = \underline{\underline{A_0 \odot B_0}}$$



$A_0 = 1 \quad B_0 = 0$

$\underline{A > B} \quad \underline{G=1}$

$A_0 = 0 \quad B_0 = 1$

$\underline{A < B} \quad \underline{L=1}$

$A_0 = B_0 = 0$ OR

$\underline{A_0 = B_0 = 1}$

$\underline{A = B} \quad \underline{E=1}$

comparators2 bit

$A \rightarrow A_1, A_0$

$B \rightarrow B_1, B_0$

$A > B ; G=1$

$A < B ; L=1$

$A = B ; E=1$

	A_1	A_0	B_1	B_0	G	L	E
0	0	0	0	0	0	0	1✓
1	0	0	0	1	0	1✓	0
2	0	0	1	0	0	1✓	0
3	0	0	1	1	0	1✓	0
4	0	1	0	0	1	0	0
5	0	1	0	1	0	0	1✓
6	0	1	1	0	0	1✓	0
7	0	1	1	1	0	1✓	0
8	1	0	0	0	1	0	0
9	1	0	0	1	1	0	0
10	1	0	1	0	0	0	1✓
11	1	0	1	1	0	1✓	0
12	1	1	0	0	1	0	0
13	1	1	0	1	1	0	0
14	1	1	1	0	1	0	0
15	1	1	1	1	0	0	1✓

A_1, A_0 \ B_1, B_0	$\overline{B_1} \overline{B_0}$	$\overline{B_1} B_0$	$B_1 \overline{B_0}$	$B_1 B_0$
$\overline{A_1} \overline{A_0}$				
$\overline{A_1} A_0$		1		3
$A_1 \overline{A_0}$	4		5	7
$A_1 A_0$	12		13	15

(Karnaugh map for G: 1s at (0,1), (1,3), (2,4), (3,7), (4,12), (5,15))

 G

A_1, A_0 \ B_1, B_0	$\overline{B_1} \overline{B_0}$	$\overline{B_1} B_0$	$B_1 \overline{B_0}$	$B_1 B_0$
$\overline{A_1} \overline{A_0}$		1		
$\overline{A_1} A_0$			1	1
$A_1 \overline{A_0}$			1	1
$A_1 A_0$				

(Karnaugh map for L: 1s at (0,1), (1,2), (1,3), (2,3))

 L

$$G = A_1 \bar{B}_1 + \bar{A}_1 \bar{A}_0 \bar{B}_1 \bar{B}_0 + A_1 \bar{A}_0 \bar{B}_1 \bar{B}_0$$

$$= A_1 \bar{B}_1 + A_0 \bar{B}_0 (\bar{A}_1 \bar{B}_1 + A_1 B_1)$$

$$G = A_1 \bar{B}_1 + \underline{A_0 \bar{B}_0} (A_1 \odot B_1)$$

$$L = \bar{A}_1 B_1 + \bar{A}_1 \bar{A}_0 \bar{B}_1 B_0 + A_1 \bar{A}_0 B_1 B_0$$

$$= \bar{A}_1 B_1 + \bar{A}_0 B_0 (\bar{A}_1 \bar{B}_1 + A_1 B_1)$$

$$L = \bar{A}_1 B_1 + \bar{A}_0 B_0 (A_1 \odot B_1)$$

$$E = \bar{A}_1 \bar{A}_0 \bar{B}_1 \bar{B}_0 + \bar{A}_1 A_0 \bar{B}_1 B_0 + A_1 \bar{A}_0 \bar{B}_1 B_0$$

$$+ A_1 \bar{A}_0 B_1 \bar{B}_0$$

$$= \bar{A}_1 \bar{B}_1 (\bar{A}_0 \bar{B}_0 + A_0 B_0) +$$

$$A_1 B_1 (A_0 B_0 + \bar{A}_0 \bar{B}_0)$$

$$= \bar{A}_1 \bar{B}_1 (A_0 \odot B_0) + A_1 B_1 (A_0 \odot B_0)$$

$$= (A_0 \odot B_0) (\bar{A}_1 \bar{B}_1 + A_1 B_1)$$

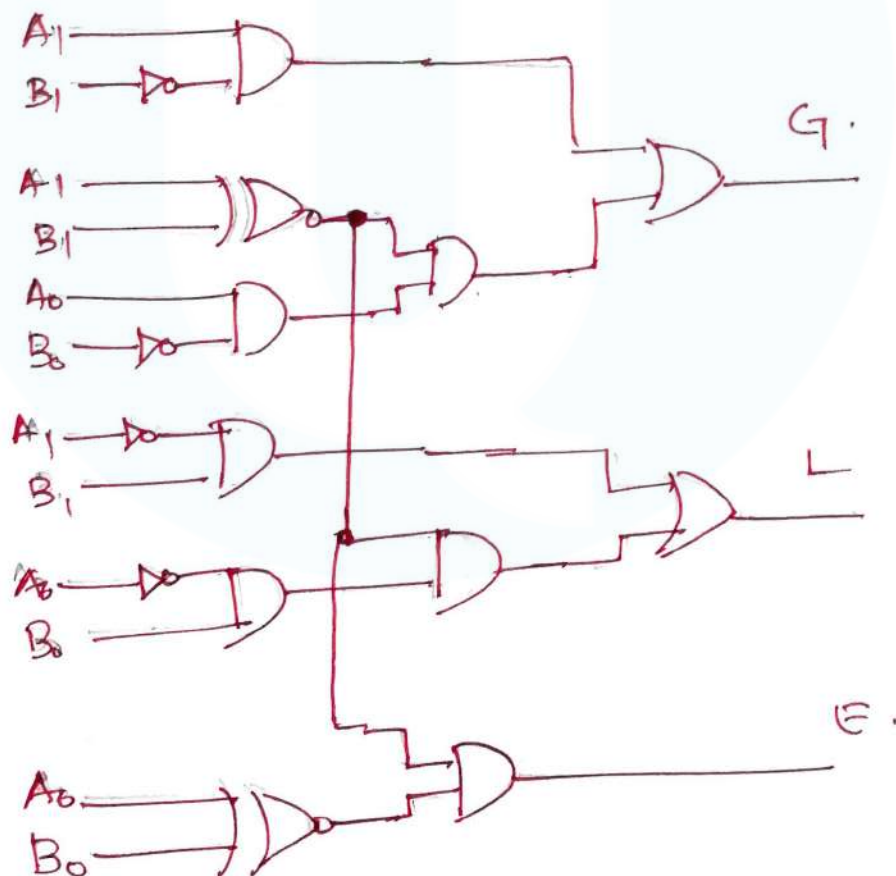
$$E = \underline{(A_0 \odot B_0)} (A_1 \odot B_1)$$

$A_1, A_0 \backslash B_1, B_0$		$\overline{B_1}\overline{B_0}$	$\overline{B_1}B_0$	$B_1\overline{B_0}$	B_1B_0
$\overline{A_1}\overline{A_0}$	0	1	2	3	4
$\overline{A_1}A_0$	5	6	7	8	9
$A_1\overline{A_0}$	10	11	12	13	14
A_1A_0	15	16	17	18	19

$$\checkmark G = A_1 \overline{B_1} + A_0 \overline{B_0} (A_1 \odot B_1)$$

$$\checkmark L = \overline{A_1} B_1 + \overline{A_0} B_0 (A_1 \oplus B_1)$$

$$E = (A_0 \odot B_0) (A_1 \odot B_1) \quad \begin{matrix} A_1, A_0 \\ B_1, B_0 \end{matrix}$$



3-bit Comparator

eg:-1

$$A \begin{cases} A_2 & A_1 & A_0 \\ 8 & 5 & 7 \end{cases}$$

$$B \begin{cases} B_2 & B_1 & B_0 \\ 6 & 8 & 3 \end{cases}$$

 $A > B$

eg:- 3

$$A \begin{cases} A_2 & A_1 & A_0 \\ 8 & 5 & 7 \end{cases}$$

$$B \begin{cases} B_2 & B_1 & B_0 \\ 8 & 5 & 4 \end{cases}$$

 $A > B$

eg:-2

$$A \begin{cases} A_2 & A_1 & A_0 \\ 8 & 5 & 7 \end{cases}$$

$$B \begin{cases} B_2 & B_1 & B_0 \\ 8 & 2 & 4 \end{cases}$$

 $A > B$

eg:- 4

$$A \begin{cases} A_2 & A_1 & A_0 \\ 8 & 5 & 7 \end{cases}$$

$$B \begin{cases} B_2 & B_1 & B_0 \\ 8 & 5 & 7 \end{cases}$$

 $A = B$ 